Architecture (ex. ARC)

Exercises Computer Architecture

1 Architecture

1.1 Stack-Architecture

Evaluate the expression $\frac{(a+bc)}{(a+dc-e)}$ using a stack-based processor evaluation stack.

a) Write pseudo code of the calculation.

b) How many direct and indirect memory references are needed in case of an infinite stacksize?

c) How many direct and indirect memory references are needed in case of a stacksize of 2?

1.2 Stack-Architecture

Evaluate the expression $\frac{(a+b)^2}{\pi} * (a + b + c)$ using a stack-based processor evaluation stack.

a) Write pseudo code of the calculation.

b) How many direct and indirect memory references are needed in case of a stacksize of 4?

c) How many direct and indirect memory references are needed in case of a stacksize of 3?

d) How many direct and indirect memory references are needed in case of a stacksize of 2?